

Defect calculus applied to toroidal sectors

Working increment v0.1 — 11 September 2026

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Status: formal application of Typed Defects v0.2 to objects already declared in TOROIDAL Working Master / Geometry Master §§7–9.

Does not alter: TD-COS-FH-001 all-orders theorem, L=3 PASS, original-pilot UNRESOLVED, physical Q2 NOT_RUN, chemistry freeze, Hidden Quotient v1.7, 39-screen algebra.

0 Declared objects (do not mix)

Fix cubic ($L \geq 2$), the ideal cosine kernel TD-COS-FH-001 at ($J=1$), ($t=1/2$), ($h_6=0$), fixed-reference ($Z_{\{000\}}$), unbounded integer currents.

Name Object

Source domain (D) Constraint-satisfying states ($X=(I,M,q)$) on the (L^3) lattice: ($\operatorname{div} I=0$), membrane constraint

Kernel (K) Attempted-microtick mixture of Geometry Master §8; identities and rejections both advance the clock

Spacing ($d \in \mathbb{Z}_{>0}$) Attempted-microtick stride; ($R=K^d$)

Sector map (π_q) ($\pi_q(X)=q \in (\mathbb{Z}/2)^3$)

Signed winding (W) ($W_\alpha=F_\alpha(k)$), integer, cut-independent on this divergence-free domain

Membrane (M) Mod-two 2-chain

Full state (X) itself; (K) is Markov on (D)

Sector process ($Y_n=\pi_q(X_{\{dn\}})$)

History-(m) record ($r_m=(Y_0, \dots, Y_{m-1})$)

Belief state ($\nu_n(\cdot \mid Y_0, \dots, Y_n)$), the causal posterior on (D)

The L=3 rotor/gauge follow-up is a different kernel (angle proposals then link flips; accepted-sweep clock). Lemma D4 applies: a result on that domain is not a result on TD-COS-FH-001.

1 Static loci

1.1 Sector from winding (coarsening)

On this source-free divergence-free domain, (q) is the parity of the integer cut class. Write

[
 $\pi_q \coloneqq \operatorname{parity} \circ W$.
]

Lemma T1 (D2). ($N_{\pi_q}(\pi_W)=\varnothing$).

Parity is a function of signed winding. The coarser label always descends through the finer one.

Lemma T2 (D2 converse fails). ($N_W(\pi_q)=(\mathbb{Z}/2)^3$) on this domain, for the natural range of (W).

Each sector fiber contains currents of every compatible even (or odd) winding of arbitrarily large ($|W|$). Distinct signed windings share a sector. Recovering load-like (W) from sector (q) fails everywhere. This is Kind I: the relation (parity / sector) survives; the specified output (signed winding) does not.

Do not identify (q) with (W) . Geometry Master already forbids replacing signed (W) by a volume average or by parity when sources are present. Here sources are zero and the identification still fails in the reverse direction.

1.2 Product of static witnesses (D1)

[
 $N_{\{(q,W)\}}(\pi_q) = N_q(\pi_q) \cup N_W(\pi_q) = N_W(\pi_q),$
]

since $(N_q(\pi_q) = \text{varnothing})$ tautologically. Retaining only (q) loses (W) . Retaining $((q,W))$ kills the static Kind I defect for (W) , and creates a different predictive question. The all-orders theorem is not a theorem about $((q,W))$. Geometry Master §8 states this limit explicitly. Defect calculus repeats it as D1: cleaning the sector witness does not clean the winding witness, and adding winding changes the predictive locus.

1.3 Membrane

(M) is independent discrete data subject to the constraint. $(N_M(\pi_q))$ is nonempty: many membranes sit over one sector. Kind IV (discrete extra sheet), not Kind I. Legal repair: retain (M) , or retain a declared functional of (M) . Illegal repair: treat sector frequencies as membrane frequencies. Dual-sector probabilities and direct holonomy counts are different observables ($L=3$ record already separates them).

1.4 Sourced restriction (D4)

With charge-six sources, integer winding is unavailable and cuts agree only modulo six.

Restricting (D) to sourced states removes the integer- (W) witness rather than repairing it. That is domain change, not descent. The modular class is a different witness. Kind III only if a stated subgroup of states is shown to inject into a stated target; do not call modular reduction “GHS” or a speedup.

2 Predictive loci (the actual theorem)

The kernel is stochastic. Use the Working Master form of D5: autonomous retained dynamics iff equal retained labels imply equal next-observation laws for every measurable target.

2.1 Full state

Lemma T3. $(N_{\{\text{Law}\}(X_1 \mid X_0)}(\text{id}_D) = \text{varnothing})$.

(K) is Markov on (D) . The full state is a clean predictive record. This is not new; it is the standing fact that the all-orders result is about a quotient, not about (X) .

2.2 One-step sector

Let $(W^{\rightarrow}_q(X) = \text{Law}(\pi_q(X_1 \mid X_0 = X))$, a probability on the eight sectors.

Lemma T4. $(N_{\{W^{\rightarrow}_q\}}(\pi_q) \neq \text{varnothing})$.

Different currents in one sector produce different next-sector laws. The proof engine of TD-COS-FH-001 is a quantitative form of this: states with current $(2N)$ on every positive edge change the return law through a Bessel-tail term as $(N \rightarrow \infty)$, so one sector label does not determine the next-sector kernel. Kind I again, now with witness = next-sector law rather than signed winding.

2.3 Finite sector history

Let (r_m) be the length- (m) sector string, $(m < \infty)$. Let $(W^{\rightarrow}_{\{q,m\}})$ be the law of the next sector given the last (m) sectors and the hidden current/membrane.

Lemma T5. For every fixed $(L \geq 2)$ and every fixed attempted-microtick spacing $(d \geq 1)$, and every finite (m) ,

[
 $N_{\{W^{\rightarrow}_{q,m}\}(r_m)} \neq \varnothing$.
]

This is the all-orders theorem, restated as “the predictive non-descent locus of every finite sector history is nonempty.” Finite Markov order would be $(N_{\{W^{\rightarrow}_{q,m}\}(r_m)} = \varnothing)$ for some (m) . The theorem says that set is never empty.

Classification: predictive Kind I, with optional Kind IV on the discrete eight-point base. It is not Kind V. The missing information (current, membrane) lies inside the (π_q) -fibers as extra source coordinates, not outside $(\operatorname{im} \pi_q)$. Sector labels are attained. The fiber is heterogeneous for the future-sector law.

2.4 Canonical predictive refinement

The causal posterior (ν_n) is (π_W) for the predictive witness “future sector path.” By Theorem 14.3 / D5, (N) of that refinement is empty: (ν_n) is a sufficient belief state. Geometry Master already says its representation may be unbounded. That is an approximation problem, not a descent problem.

Do not feed future sectors into (ν_n) . The filter is a function of the observed past and the kernel.

2.5 Horizon diameters

$(\delta_h^q(z))$ on a sector-history fiber is nondecreasing in (h) . The all-orders argument is stronger than one-step: it produces a defect that survives every finite memory. A single tested pair of currents in one sector would only lower-bound (δ_1) . The Bessel-tail family is a sequence of pairs that keeps the defect alive as (N) grows, which is why the order cannot stabilize.

2.6 What the theorem does not say (D4 / D3)

Change	Calculus	Status
Finite-current truncation	D4, smaller (D)	Not covered. Locus may shrink.
UNRESOLVED for that (D)		
Accepted-move clock different (π) / different (K)		Not covered
Adaptive schedule	different (K)	Not covered
Record $((q, W))$	D1, different witness	Not covered
$L=3$ rotor/gauge sweep	different (K) , D4	Separate PASS; not this locus
Belief-state finite compression	approximation of a clean record	Not (N) ; needs (B, h, ϵ)
Physical Q2	different ensemble, estimator, claim	NOT_RUN

3 Kind table for this kernel

Kind	Toroidal instance	Legal repair	Illegal repair
I Operation / output loss	$(q = \operatorname{parity}(W))$ loses (W) ; (q) -history loses next- (q) law		
Retain (W) , or retain (ν_n) , or retain declared current functionals	Treat sector string as winding; treat finite Markov table as the kernel		

II Unit / character No native unit group here. Character transform ($\mathcal{Z}_q \leftarrow \mathcal{Z}_h$) is an ensemble change, not a local log Keep ($\mathcal{Z}_h$) and ($\mathcal{Z}_q$) as separate witnesses Call a dual coefficient a Schirokauer map

III Subgroup Source-free (D) vs sourced (D); integer (W) only on ($\operatorname{div} l = 0$)

Restrict and rename the witness (modular class) Infer sourced winding from source-free formulas

IV Discrete sector Eight-point base; membrane sheets over one (q); holonomy 8-bin counts

Retain sheet / membrane / holonomy label as its own witness Treat ($\hat{p}(q)$) counts as ($\hat{p}(q)$)

V Representability Finite-current or finite-history encodings that miss functionals of unbounded (I) Name the omitted functional; enlarge observation Call “no finite Markov order” a missing dual. It is a fiber collision

Kind II is listed to forbid a false identification with the NFS packet, not because a unit character is required on this kernel.

4 Transport, holonomy, shape

($Q = (\mathbb{Z}/2)^3$) is a discrete eight-point set. Under the standing topology of (Q), ($N_{\{W \rightarrow \pi_q\}}(\pi_q)$) is a subset of eight points. Openness, (∂N), and continuous holonomy do not apply to this (Q) (no-shape + missing hypotheses of Thm 14.5).

Witness-holonomy in the sense of Prop. 14.8 would require a declared connection on the fibers ($F_q = \{X: \pi_q(X) = q\}$) and loops in a continuous base. The discrete sector walk is not that object.

Direct holonomy ($h \in (\mathbb{Z}/2)^3$) in the L=3 record is a different witness on a different kernel. Nonzero ($\frac{h}{W}$) would imply non-descent if those hypotheses were met; the converse remains false.

To speak of ($\partial N_{\{W, \varepsilon\}}$) one must enlarge the base to a parameter space (e.g. ((J, t, L))) on which (δ) is defined and continuous. That enlargement has not been declared.

Do not draw a boundary on the eight-point cube and call it lithium.

Ensemble note, D1 again: ($\mathcal{Z}_h$) and ($\mathcal{Z}_q$) are different witnesses related by a character transform. ($\mathcal{Z}_{\operatorname{full}} = \mathcal{Z}_{\{000\}}$) while ($\mathcal{Z}_{\{000\}} = \frac{1}{18} \sum_q \mathcal{Z}_q$). Full periodic summation forcing even-parity projection is an algebraic projection, not observed confinement, and not ($N_{\{\pi_q\}}$).

5 Clock

Attempted-microtick time is part of (K). Switching to accepted-move time is a different record. If fixed-time sector laws agreed and cross-time laws did not, a clock coordinate would be the Working Master repair. The all-orders theorem already fails inside one clock (attempted microticks). Adding a clock cannot repair a same-time fiber collision (Working Master §3). The defect is not a missing clock.

6 Empirical gate for this application

Frozen tuple for any numerical sector run that wants a defect-calculus readout:

(D): named kernel, (L), current bound or “unbounded,” source convention

(π): (q), or (r_m), or ((q,W)), named

(W): next-sector law, or signed (W), or dual ($\hat{p}(q)$), named

(d_Y): total variation on sector measures, or integer distance on (W)

(\varepsilon), coverage, same-fiber rule

Defect kind: I (predictive) for the all-orders claim; I (static) for winding-from-sector; IV for membrane/holonomy mixups

Certified predictive non-descent on a truncated current box is only a lower bound on that box (D4). It does not replay the Bessel-tail argument and does not inherit the all-orders label.

L=3 PASS remains a separate engineering screen on holonomy counts and dual-reference probabilities. It is not a measurement of $(N_{W \rightarrow q}(\pi_q))$ for TD-COS-FH-001.

7 Disposition

The sector theorem is the statement that every finite sector history has nonempty predictive non-descent locus for next-sector law, under the declared kernel and attempted-microtick clock.

[
\boxed{\text{SECTOR THEOREM = PREDICTIVE } N_{\neq} \text{ FOR EVERY FINITE } r_m}

]
[
\boxed{\text{STATIC } N_W(\pi_q) \text{ NONEMPTY; } N_q(\pi_W) \text{ EMPTY}}

]
[
\boxed{\text{BELIEF STATE } = \text{ CANONICAL PREDICTIVE REFINEMENT; COMPRESSION SEPARATE}}

]
[
\boxed{\text{ALL SOURCE STATUSES UNCHANGED}}

]
Point TOROIDAL Working Master at this increment for the typing. Do not paste the typing into the all-orders proof. The proof stays the proof.